

3 Fruit flies of the genus *Bactrocera* are pests that damage fruit crops.

Table 3.1 gives the names and geographical ranges of *Bactrocera* fruit flies that were classified as four separate species.

Table 3.1

name	geographical range
<i>B. dorsalis</i>	China, India, Thailand
<i>B. invadens</i>	Africa, India
<i>B. papayae</i>	Indonesia, Malaysia
<i>B. philippinensis</i>	Philippine Islands, Borneo

In 2014, the classification of these flies was changed. All four species were recognised as belonging to a single species, *B. dorsalis*.

(a) (i) Suggest reasons why the four species were originally classified as separate species.

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(ii) Some students decided to investigate whether the flies were members of one species or four separate species.

Suggest a simple investigation that the students could carry out.

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- (b) Female *B. dorsalis* lay their eggs in different types of fruits, such as avocados, bananas, mangos and papayas. The eggs hatch into larvae that eat the fruit. The actions of the female flies and larvae allow microorganisms to enter the fruit. The microorganisms feed by secreting extracellular enzymes, causing the fruit to rot.

Name **two** kingdoms that include organisms that could spoil fruit by secreting extracellular enzymes.

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Before the four *Bactrocera* species were reclassified as a single species in 2014, some governments in Asia banned fruit imports from African countries to avoid introducing *B. invadens* as an alien species.

- (c) Explain why the introduction of alien species should be avoided.

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- (d) Suggest how reclassifying *B. invadens* as *B. dorsalis* will benefit fruit-producing countries in Africa.

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